

Code-Layout, Readability and Reusability

Date	30 September 2026
Team ID	SWTID-2026-6111
Project Name	FIT BUDDY AI
Maximum Marks	55 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Partial	Indentation style is not explicitly documented; code samples are structured.
2	Proper File Structure	Files and folders are logically organized	Yes	Separate app, templates and static folders are clearly defined.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Descriptive fields and variables such as workout_plan and user inputs are documented.
4	Function / Method Names	Functions are descriptively named	Yes	Functions are descriptively named, e.g. generate_workout_gemini().
5	Code Comments	Inline and block comments explain logic	Partial	Inline/block code comments are not explicitly described in the project document.
6	Modular Design	Code is split into reusable functions/modules	Yes	AI, database, routing and template functions are separated into modules.
7	No Redundant Code	Duplicate or unused code is removed	Partial	Modularity is documented, but duplicate/unused-code checks are not explicitly stated.
8	Error Handling	Exceptions and errors are handled gracefully	Partially	Error handling is not specifically documented in the supplied project document.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technolog	yWhere Reused	Reusability Level (High / Medium / Low)
1	generate_workout_gemini()	Python + Gemini 1.5 Pro	routes.py / result.html	High

2	generate_nutrition_tip_with_flash()	Python + Gemini Flash	routes.py / result.html	High
3	update_workout_plan()	Python + Gemini 1.5 Pro	/submit-feedback / updated_plan.py	High
4	Database functions: save_user(), save_plan(), update_plan(), get_user()	Python + SQLAlchemy + SQLite	routes.py and admin data access	Medium
5	Jinja2 templates: index.html, result.html, all_users.html	HTML + CSS + Jinja2	FastAPI-rendered frontend pages	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	4	Modular FastAPI structure with separate routing, AI, database and template layers.
Readability	4	Descriptive functions, clear workflow and structured template/data binding are documented.
Reusability	5	Multiple reusable AI and database functions are separated into dedicated modules.
Documentation / Comments	4	Project documentation is detailed; inline code comments are not explicitly documented.
Overall Score	4	Strong overall structure and modularity, with some code-quality areas not explicitly verified.